GURUDUTT — 3-Minute College Presentation Script

Territory Runner | Subsystem Focus: UI/UX Design & Athletic Ergonomics

`TOTAL TIME: 3:00 MIN`

ROLE: UI/UX LEAD

PRESENTER	ASSIGNED TIME	CORE STACK
GURUDUTT	Minute 3:00 to 6:00 (3 Mins)	Flutter 3.13+, Ink + Blue, Map HUD

⏱️ MINUTE 1 (3:00 – 4:00) : Athletic Ergonomics & Design Philosophy

`3:00 - 4:00`

"Good morning respected examiners. My name is **Gurudutt**, and in *Territory Runner*, I led the **UI/UX design, visual design system, tactical map interface, and athletic ergonomics** built with Flutter.

When an athlete runs outdoors, they face extreme visual and physical constraints that break typical app designs: **harsh direct sunlight** washes out washed-out pastel screens, and **physical bouncing and sweaty fingers** make standard small buttons impossible to tap reliably.

To solve this, I designed our bespoke **Athletic Minimal Design System: Ink and Pacer Blue**. We rejected generic white backgrounds and colorful arcade clutter, building a calm, high-contrast tactical dark interface inspired by aviation heads-up displays."

> 👉 ***Action:*** *Switch to Slide 5. Display the high-contrast Ink & Pacer Blue color tokens and running HUD mockup.*

⏱️ MINUTE 2 (4:00 – 5:00) : Tactical Map HUD & 1400ms Hold Gesture

`4:00 - 5:00`

"Our color architecture is built around three calibrated tokens: a **deep carbon canvas (`#090B0F`)** which eliminates screen glare and minimizes OLED battery consumption; crisp **Pacer Blue (`#4C8DFF`)** for sovereign athlete trails and conquered sectors; and **Coral (`#FF7A59`)** for contested rival boundaries.

On the **Tactical Map HUD**, dynamic vector polylines and polygons update continuously at **60 FPS**. I isolated rendering layers using Flutter `RepaintBoundary` widgets, preventing heavy map tile re-renders on every GPS tick.

To solve the sweaty-finger problem, I eliminated the standard 'Finish' button. In outdoor testing, runners frequently stop runs by mistake. I created a circular **1400-millisecond Hold-to-Finish gesture** with intermediate haptic feedback pulses every 350 milliseconds, requiring deliberate physical intent."

> 👉 ***Action:*** *Switch to Slide 6. Show the 1400ms Hold-to-Finish circular animation and map layer diagram.*

⏱️ MINUTE 3 (5:00 – 6:00) : AppToast Framework, Metrics & Handover

`5:00 - 6:00`

"For in-run athlete feedback, I built **AppToast**—a unified notification overlay that alerts runners to GPS signal degradation, rival territory contestation, and server area reconciliations without obstructing the map viewport.

The UI operates on zero raster fallbacks, using vector SVGs and typography for razor-sharp rendering on any screen density. The mobile UI is validated across **317 automated Flutter tests** with zero layout overflow errors and clean Dart static analysis.

Now, I pass the floor to my teammate **Avinash**, who will present our computational geometry algorithms, loop detection, and project documentation."

> 👉 ***Action:*** *Switch to Slide 7. Show the 317 Flutter tests passing badge and nod toward Avinash.*

GURUDUTT — Slide Deck Presentation Guide

Exactly what to display and point at during your 3 minutes

4 SLIDES TOTAL

SLIDE 5 (3:00 - 3:45) DESIGN PHILOSOPHY

Athletic Minimal Design System

- Direct sunlight glare vs high-contrast tactical dark interface.
- **Ink Canvas (`#090B0F`):** Zero glare, maximum OLED battery savings.
- **Pacer Blue (`#4C8DFF`):** Crisp high-visibility athletic sovereign color.
- **Coral (`#FF7A59`):** High-alert rival indicator.

SLIDE 6 (3:45 - 4:30) TACTICAL MAP HUD

60 FPS Map Layers & Vector Rendering

- Multi-layer architecture: ArcGIS Dark Tiles → Polygons → Trails → HUD.
- `RepaintBoundary` isolation preventing full map widget rebuilds.
- Smooth runner marker heading rotation and position interpolation.

SLIDE 7 (4:30 - 5:15) ATHLETIC ERGONOMICS

1400ms Hold-to-Finish & AppToast

- Accidental touch prevention: sweat and physical bouncing mitigation.
- 1400ms circular hold gesture with haptic vibration feedback.
- Unified AppToast framework for server area reconciliation alerts.

SLIDE 8 (5:15 - 6:00) UI VERIFICATION

Zero Raster Fallback & 317 Tests

- Pure vector SVGs & typography across 320dpi to 560dpi mobile screens.
- **317 Automated Flutter Tests Passing** with 0 analyze warnings.
- Responsive HUD adapting cleanly across Android and iOS notch layouts.

📋 Gurudutt's Quick Reference Pocket Card

CODE & FILES

- **Theme Tokens:** `lib/core/theme/` (Carbon `#090B0F`, Pacer Blue `#4C8DFF`)
- **Tactical Map Layers:** `lib/features/map/widgets/map_layers.dart`
- **Hold-to-Finish Logic:** `lib/features/gameplay/territory_service.dart`
- **Notification Toasts:** `lib/ui/widgets/app_toast.dart`
- **Key Gesture:** 1400ms continuous contact duration + 350ms haptic ticks.

GURUDUTT — Examiner Viva Defense Cheat Sheet

Top 4 killer examiner questions with rapid 15-second oral defenses

EXAMINER DEFENSE

Q1: Why did you build a custom dark theme instead of using standard Material 3 Light Theme?

15-Second Defense: "Athletic usability: white and pastel screens wash out completely in direct noon sunlight and cause severe pupil glare. Furthermore, running GPS apps consume significant battery; our Ink `#090B0F` canvas turns off OLED pixels on modern AMOLED displays, reducing screen power drain by up to 35% during long outdoor runs."

📁 Implemented in: `lib/core/theme/app_theme.dart`

Q2: How does the 1400ms hold gesture prevent accidental input during exercise?

15-Second Defense: "Biomechanical tremor: runners in motion with sweaty fingers frequently register accidental double-taps or pocket-rubs on standard modal dialogs. Our 1400ms hold gesture requires continuous finger contact, animates an SVG circular progress ring, and pulses haptic ticks every 350ms, confirming deliberate physical intent without stopping their run."

📁 Implemented in: `lib/features/gameplay/territory_service.dart`

Q3: How do you keep the tactical map rendering at 60 FPS without frame stutter during live GPS updates?

15-Second Defense: "By using isolated Flutter `RepaintBoundary` widgets. The static map tiles, sovereign territory polygons, and live breadcrumb trail are decoupled into separate CustomPainters. When a new GPS waypoint arrives at 1 Hz, Flutter repaints only the dynamic breadcrumb vector layer while reusing cached raster tile GPU textures."

📁 Implemented in: `lib/features/map/widgets/map_layers.dart`

Q4: How does the UI handle area reconciliation if the server's PostGIS area differs from the local preview?

15-Second Defense: "Via our non-intrusive AppToast notification framework. The local preview displays an instant optimistic polygon. When the HTTP 201 response returns the authoritative PostGIS area, if the variance exceeds 0.5%, an AppToast gently reconciles the metric on screen without interrupting the athlete's workout summary."

📁 Implemented in: `lib/ui/widgets/app_toast.dart`

Candidate Signature	Internal Project Guide	External Viva Examiner
GURUDUTT	Dept. of CSE	University Board